

Title

Cluster-randomized trial of structured breastfeeding support to promote lactation continuation and reduce socioeconomic disparities: a mixed-methods study protocol from Scandinavia

Abstract

Breastfeeding is foundational for child health outcomes, yet rates remain suboptimal and unequally distributed across educational and socioeconomic backgrounds in high-income Nordic countries. This protocol describes a cluster-randomized trial evaluating an evidence-based breastfeeding support program designed to extend exclusive breastfeeding duration and mitigate educational and economic disparities in lactation outcomes. The trial enrolls 21 municipal healthcare clusters across two Danish regions, implementing a coordinated intervention combining intensive health-visitor training, standardized family support protocols, accessible resource toolkits, and structured follow-up mechanisms over an 18-month intervention period. Primary effectiveness outcomes measure exclusive breastfeeding continuation rates and socioeconomic-stratified disparities. Secondary assessments include program fidelity audits, contextual barriers to implementation, health-visitor confidence measures, and family experiences captured through qualitative inquiry. This protocol outlines a pragmatic evaluation of scalable system-level breastfeeding promotion within existing primary healthcare infrastructure.